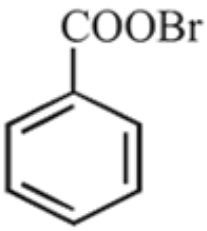
Quizrr Chapter-Wise Test for JEE Main - 2025

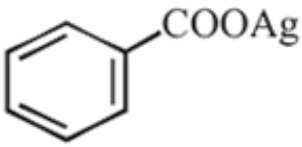
Q.1

Silver benzoate will react with bromine in CCl_4 to form:

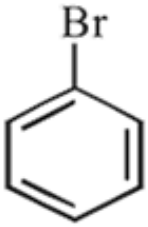
(A)



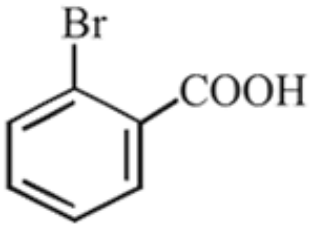
(B)



(C)



(D)



Q.2

$R - CH_2 - CH_2OH$ can be converted in $R - CH_2CH_2COOH$. The correct sequence of reagents is

(A) PBr_3, KCN, H^+

(B) PBr_3, KCN, H_2

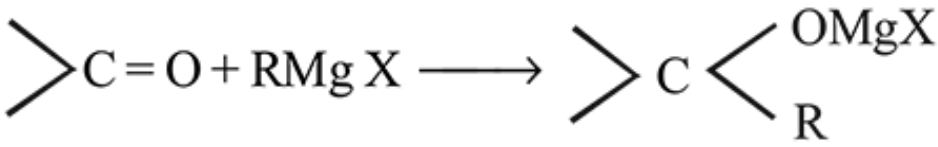
(C) KCN, H^+

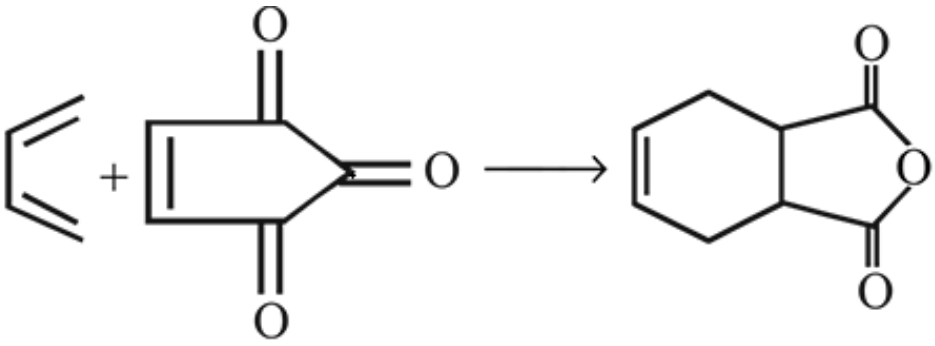
(D) HCN, PBr_3, H^+

Q.3

Which among the following is substitution reaction ?

@IITJEE_Advanced

(A) 

(B) 

(C) $R_3COH \xrightarrow[\text{Conc. HCl}]{\text{Anhy. } ZnCl_2}$ Turbidity

(D) All of these

Q.4

The reaction conditions leading to the best yields of C_2H_5Cl are:

(A) $C_2H_6(\text{ excess }) + Cl_2 \xrightarrow{\text{UV light}}$

(B) $C_2H_6 + Cl_2 \xrightarrow[\text{room temperature}]{\text{dark}}$

(C) $C_2H_6 + Cl_2 \text{ (excess)} \xrightarrow{\text{UV light}}$

(D) $C_2H_6 + Cl_2 \xrightarrow{\text{UV light}}$

Q.5

The order of reactivity of the given haloalkanes towards nucleophile is :

(A) $RI > RBr > RCl$

(B) $RCl > RBr > RI$

(C) $RBr > RCl > RI$

(D) $RBr > RI > RCl$

Q.6

$C_2H_5Br \xrightarrow[AgCN]{} Xn \xrightarrow[Zn-Hg/HCl]{\text{Reduction}} Y$, Here Y is

(A) Ethyl methyl amine

(B) n -propylamine

(C) Isopropylamine

(D) Ethylamine

@IITJEE_Advanced

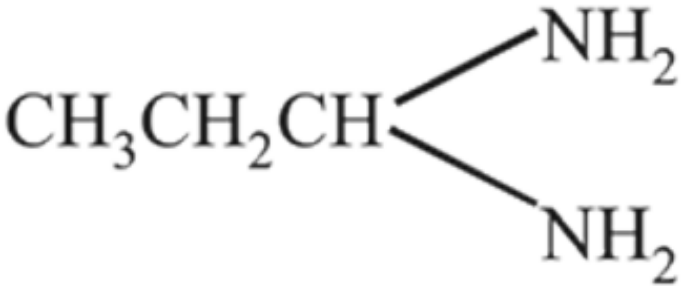
Q.7

When $CH_3CH_2CHCl_2$ is treated with $NaNH_2$, the product formed is

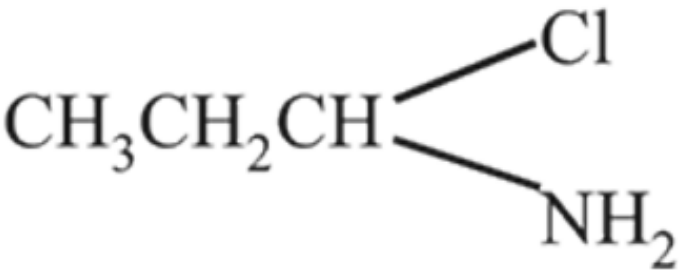
(A) $CH_3 - CH = CH_2$

(B) $CH_3 - C \equiv CH$

(C)



(D)



Q.8

$CH_3Br + Nu^- \longrightarrow CH_3 - Nu + Br$

The decreasing order of the rate of the above reaction with nucleophiles (Nu^-) A to D is $[Nu^- = (A)PhO^-, (B)AcO^-, (C)HO^-, CH_3CH_2O^-]$

(A) $A > B > C > D$

(B) $B > D > C > A$

(C) $D > C > A > B$

(D) $D > C > B > A$

Q.9

Which of the following statements is wrong?

(A) Ethyl chloride on reduction with $Zn - Cu$ couple and alcohol gives ethane.

(B) The reaction of ethyl magnesium bromide with acetone gives 2-Pentanol.

(C) Alkyl halides follow the following reactivity sequence on reaction with alkenes. $R - I > R - Br > R - Cl > R - F$

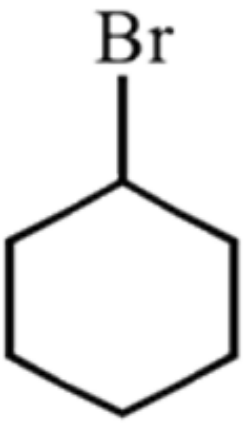
(D) $C_2H_4Cl_2$ may exist in two isomeric forms.

@IITJEE_Advanced

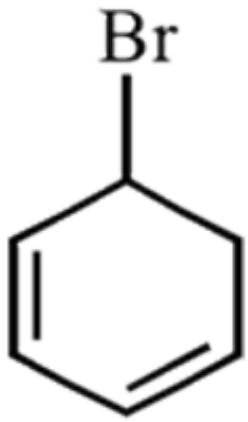
Q.10

Rate of S_N2 will be negligible in :

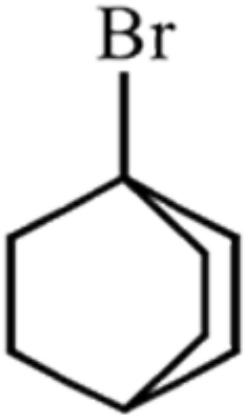
(A)



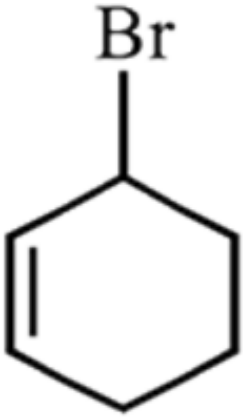
(B)



(C)



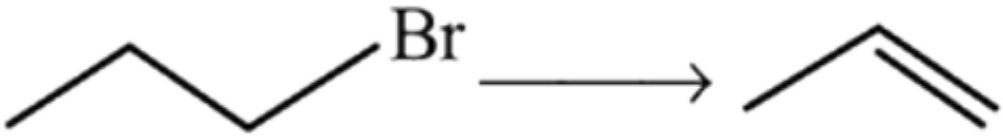
(D)



@IITJEE_Advanced

Q.11

Which one of the following reagents is not suitable for the elimination reaction ?



(A) NaI/acetone

(B) $NaOEt/EtOH$

(C) $NaOH/H_2O$

(D) $NaOH/H_2O - EtOH$

Q.12

Identify Z in the following serie's

$$C_2H_5I \xrightarrow{\text{Alc. KOH}} X \xrightarrow{Br_2} Y \xrightarrow{\text{excessKCN}} Z$$

(A) CH_3CH_2CN

(B) $NCCH_2 - CH_2CN$

(C) $BrCH_2 - CH_2CN$

(D) $BrCH = CHCN$

Q.13

Among the following, the molecule with the lowest dipole moment is

(A) $CHCl_3$

(B) CH_3Cl

(C) CH_2Cl_2

(D) CCl_4

Q.14

Which chloride is least reactive with the hydrolysis point of view?

(A) CH_3Cl

(B) CH_3CH_2Cl

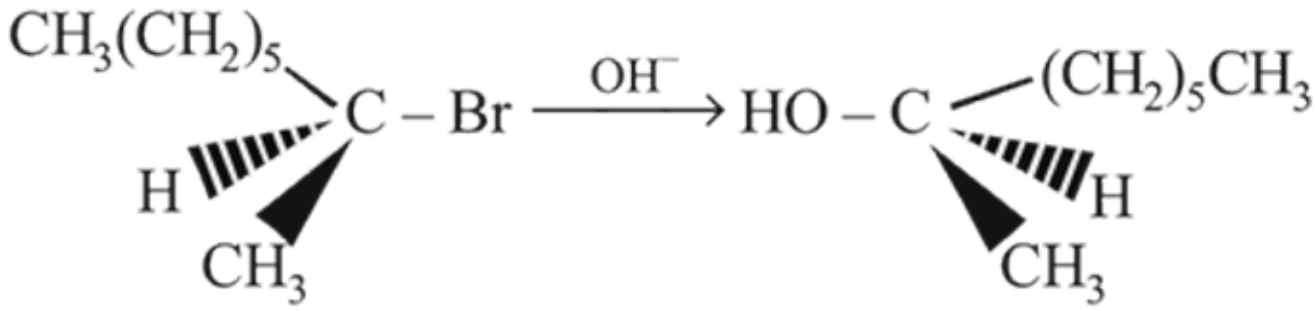
(C) $(CH_3)_3CCl$

(D) $CH_2 = CH - Cl$

@IITJEE_Advanced

Q.15

The reaction is described as



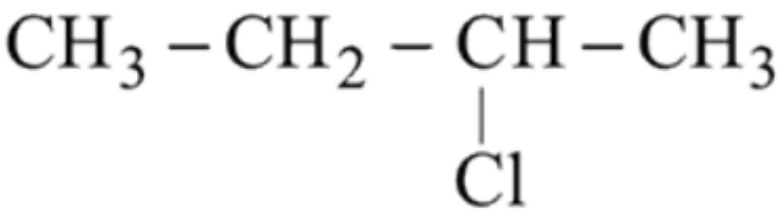
(A) S_E2

(B) S_N1

(C) S_N2

(D) S_N0

Q.16



obtained by chlorination of n -butane, will be

(A) meso-form

(B) racemic mixture

(C) d -form

(D) l -form

Q.17

2-Bromopentane is heated with potassum ethoxide in ethanol. The major product obtained is

(A) 2 -Ethoxypentane

(B) Pentene-1

(C) cis-Pentene-2

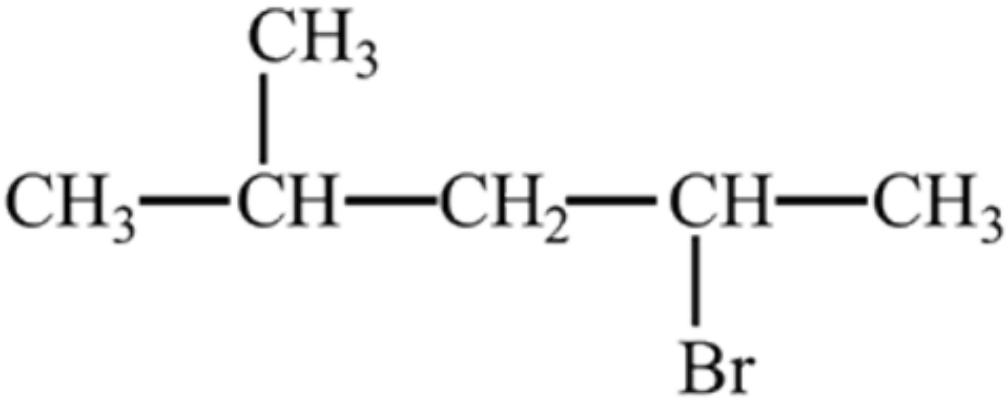
(D) trans-Pentene-2

@IITJEE_Advanced

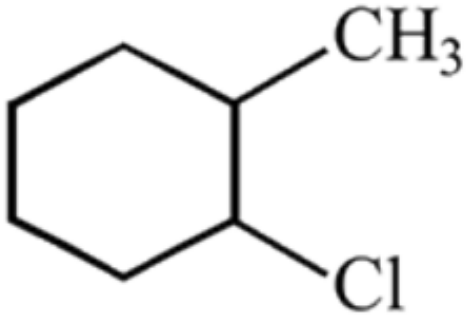
Q.18

Among the given halides, which one will give same product in both S_N1 and S_N2 reactions?

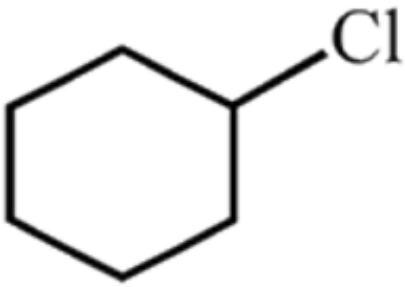
(I)



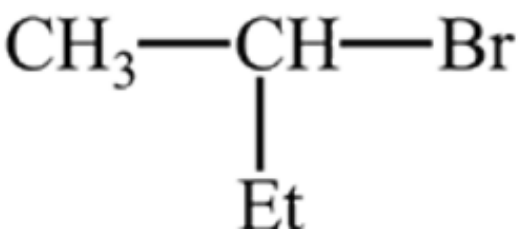
(II)



(III)



(IV)



(A) (III) only

(B) (I) and (II)

@IITJEE_Advanced

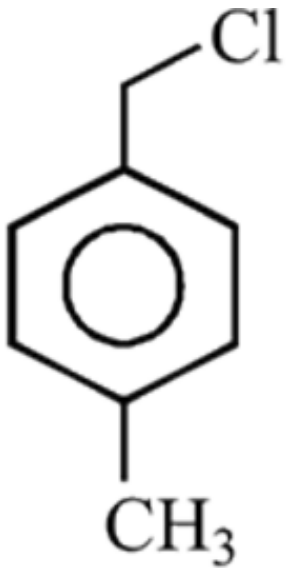
(C) (III) and (IV)

(D) (I), (III) and (IV)

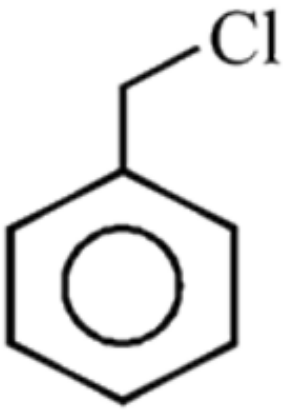
Q.19

Which of the following is most reactive towards S_N2 reaction?

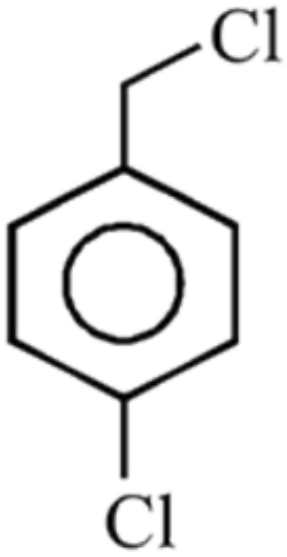
(A)



(B)

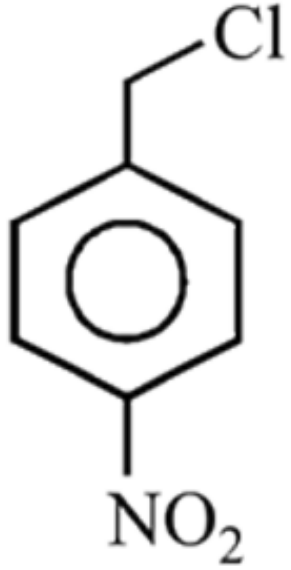


(C)



@IITJEE_Advanced

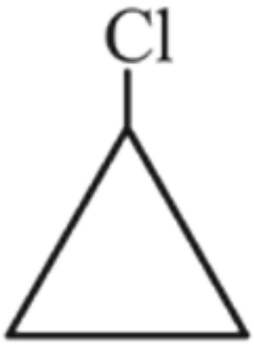
(D)



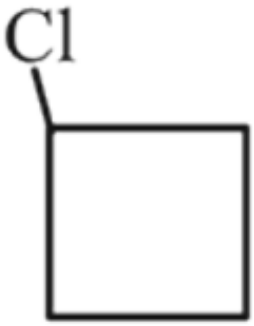
Q.20

Identify correct reactivity order for S_N1 reaction

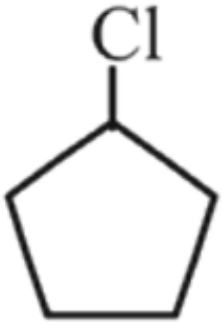
(i)



(ii)



(iii)



@IITJEE_Advanced

(A) $\mathrm{i} > \mathrm{ii} > \mathrm{iii}$

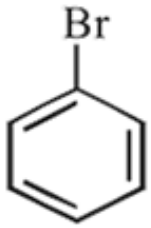
(B) $\mathrm{ii} > \mathrm{iii} > \mathrm{i}$

(C) $\mathrm{i} > \mathrm{iii} > \mathrm{ii}$

(D) $\mathrm{iii} > \mathrm{ii} > \mathrm{i}$

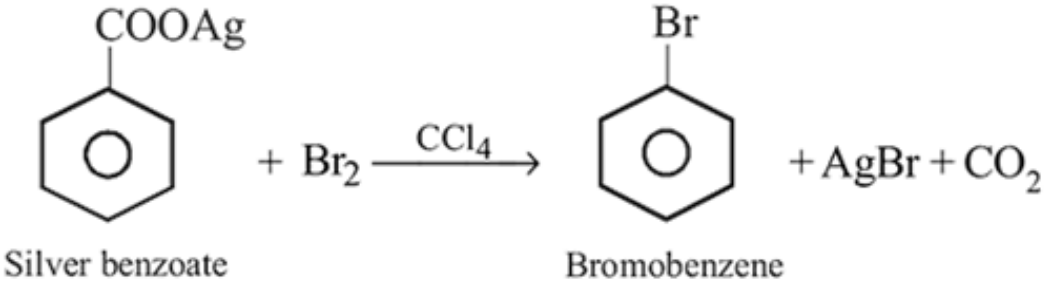
Answers & Solutions

Q.1 Answer:



Solution:

This is Hunsdiecker's reaction.



Q.2 Answer:

$\mathrm{PBr_3}, \mathrm{KCN}, \mathrm{H^+}$

Solution:

$$\mathrm{RCH_2CH_2OH} \xrightarrow{\mathrm{PBr_3}} \mathrm{RCH_2CH_2Br} \xrightarrow{\mathrm{KCN}} \mathrm{RCH_2CH_2CN} \xrightarrow{\mathrm{H_2O/H^+}} \mathrm{RCH_2CH_2COOH}$$

Q.3 Answer:

$$\mathrm{R_3COH} \xrightarrow[\text{Conc. HCl}]{\text{Anhy. } \mathrm{ZnCl_2}} \text{Turbidity}$$

Solution:

$\mathrm{R_3COH} \xrightarrow[\text{Conc. HCl}]{\text{Anhy. } \mathrm{ZnCl_2}} \mathrm{R_3C.Cl}$

It is substitution reaction. $\mathrm{R_3C.Cl}$ **being insoluble in** $\mathrm{H_2O}$ **give turbid solution.**

Q.4 Answer:

$$\mathrm{C_2H_6(\ excess\ ) + Cl_2} \xrightarrow{\text{UV light}}$$

Solution:

$$\mathrm{C_2H_6(\ \textbf{excess}\ ) + Cl_2} \xrightarrow{\text{UV light}} \mathrm{C_2H_5Cl + HCl}$$

@IITJEE_Advanced

Q.5 Answer:

$RI > RBr > RCl$

Solution:

For a given alkyl group, the order of reactivity is $\leftarrow \underset{\text{increasing bond energy}}{R-I > R-Br > R-Cl > R-F}$

decreasing halogen

reactivity.

This order depends on the carbon-halogen bond energy; the carbon-fluorine bond energy is maximum and thus fluorides are least reactive while carbon-iodine bond energy is minimum hence iodides are most reactive.

Q.6 Answer:

Ethyl methyl amine

Solution:

$$C_2H_5Br \xrightarrow{AgCN} \underset{(X)}{C_2H_5NC} + AgBr$$

$$\downarrow \text{Reduction } Zn - Hg / HCl$$

$$\underset{(Y)\text{ ethyl methyl amine}}{C_2H_5NHCH_3}$$

Q.7 Answer:

$CH_3 - C \equiv CH$

Solution:

$$CH_3 - CH_2 - CHCl_2 \xrightarrow{\frac{NaNH_2}{\Delta}}$$
$$CH_3 - CH = CHCl \xrightarrow{\frac{NaNH_2}{\Delta}} \underset{\text{Final Product}}{CH_3 - C \equiv CH}$$

Q.8 Answer:

$D > C > A > B$

@IITJEE_Advanced

Solution:

The acid character follows the order :

$CH_3COOH > C_6H_5OH > H_2O > CH_3OH$

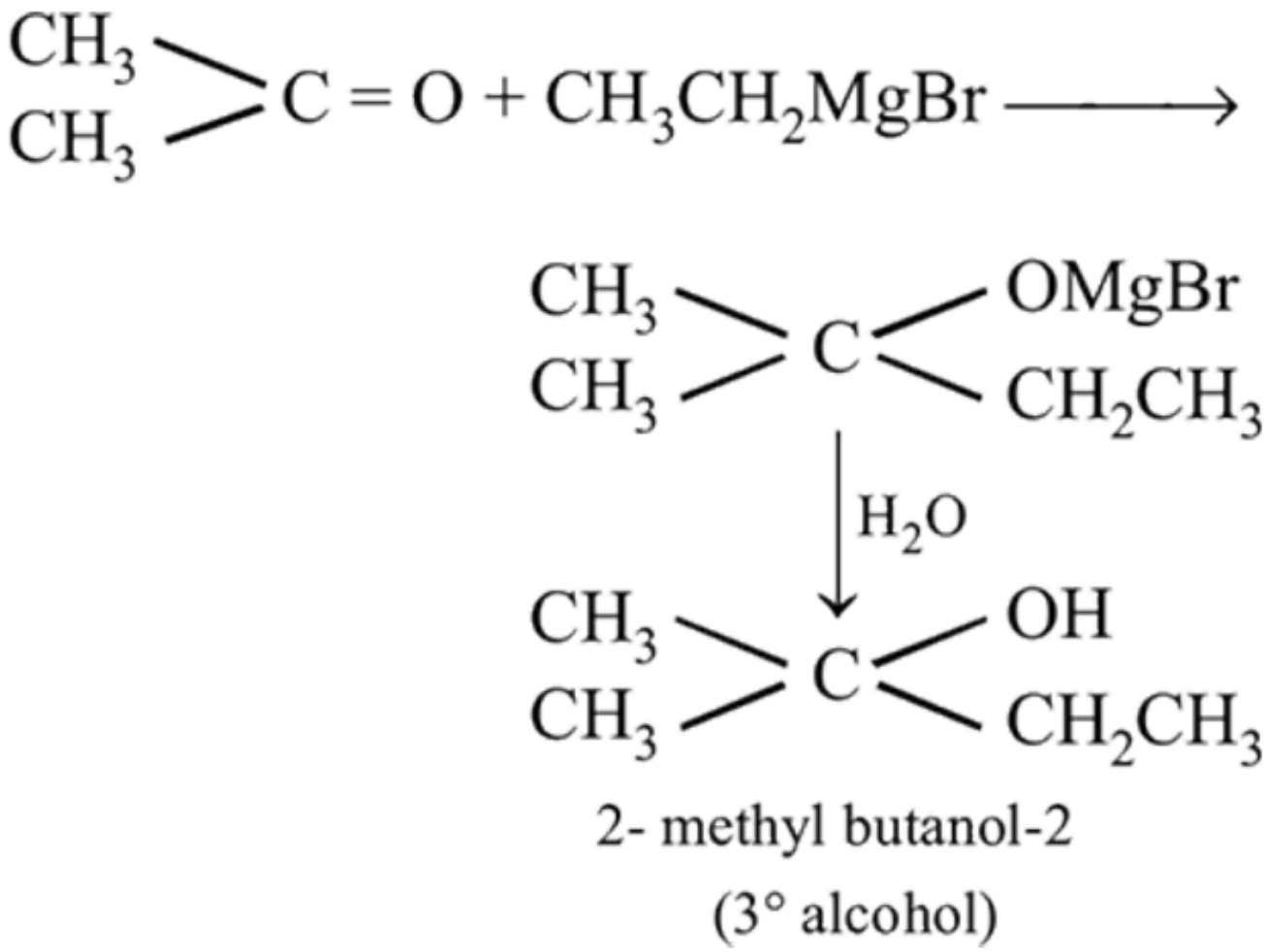
The basic character will follow the order $CH_3COO^- < C_6H_5O^- < OH^- < CH_3O^-$

The stronger the acid, the weaker the conjugate base formed.

Q.9 Answer:

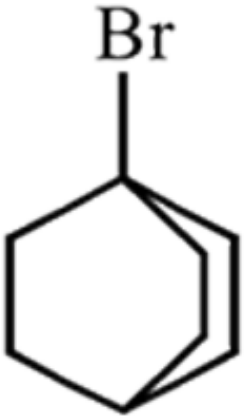
The reaction of ethyl magnesium bromide with acetone gives 2-Pentanol.

Solution:



Q.10 Answer:

@IITJEE_Advanced



Solution:

At bridge head position S_N1 and S_N2 do not takes place.

Q.11 Answer:

NaI/acetone

Solution:

Alkyl chloride or bromide undergo substitution and get converted to an alkyl iodide on treatment with a solution of sodium iodide in acetone. e.g.

$$CH_3CH_2CH_2Br + NaI \xrightarrow{\text{acetone}} CH_3CH_2CH_2I + NaBr$$

This reaction is also known as Finkelstein Reaction.

Q.12 Answer:

$NCCH_2 - CH_2CN$

Solution:

$$C_2H_5I \xrightarrow{\text{alc. KOH}} CH_2 = CH_2 \xrightarrow{Br_2}$$
$$BrCH_2 - CH_2Br \xrightarrow{\text{excessKCN}} CNCH_2 \cdot CH_2CN$$

Q.13 Answer:

CCl_4

Solution:

CCl_4 has lowest (zero) dipole moment. This is due to its symmetrical tetrahedral structure. Due to which dipole moment of one bond is cancelled by opposite dipole moment of the other three bonds.

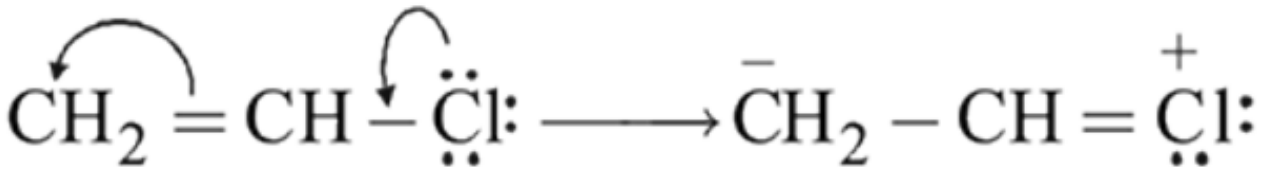
Q.14 Answer:

$CH_2 = CH - Cl$

Solution:

@IITJEE_Advanced

The non-reactivity of chlorine atom in vinyl chloride is due to resonance stabilisation.



Q.15 Answer:

S_N2

Solution:

Inversion in configuration occurs in S_N^2 reactions.

Q.16 Answer:

racemic mixture

Solution:

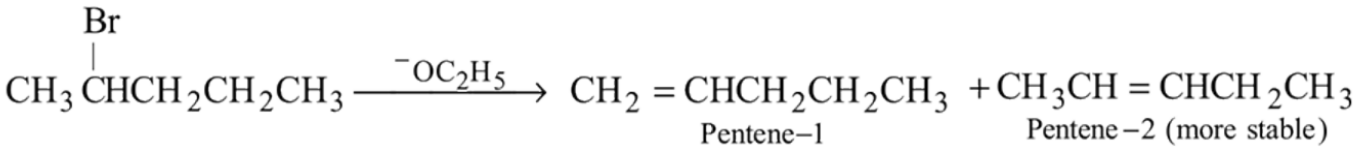
Chlorination of alkanes is a free-radical reaction. Since the intermediate free radical is planar $(sp^2$ hybridised) it can be attacked on either side of the face forming racemic mixture

Q.17 Answer:

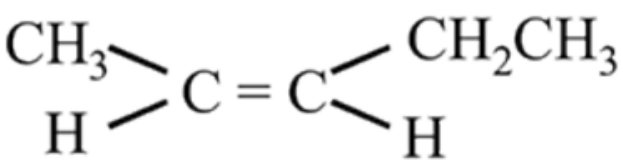
trans-Pentene-2

Solution:

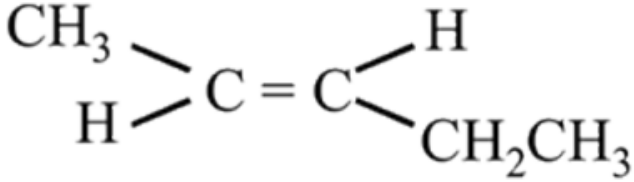
Potassium ethoxide is a strong base, hence causes elimination reaction. The alkene formed is governed by Saytzeff rule "more substituted an alkene, higher will be its stability".



Further pentene-2 shows geometrical isomerism, cis- and trans- in which trans-isomer having bulkier groups away from each other is more stable than the cis -isomer.



cis-Pentene-2



trans-Pentene-2 (more stable)

Q.18 Answer:

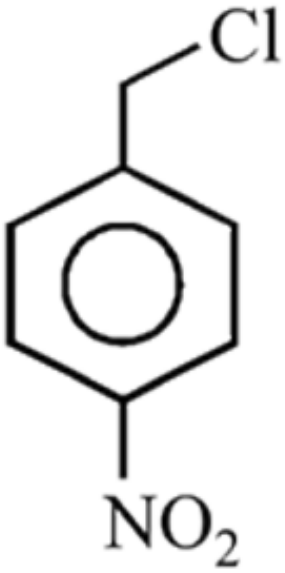
(I), (III) and (IV)

Solution:

@IITJEE_Advanced

S_N2 and S_N1 same, if C^+ not rearrange.

Q.19 Answer:



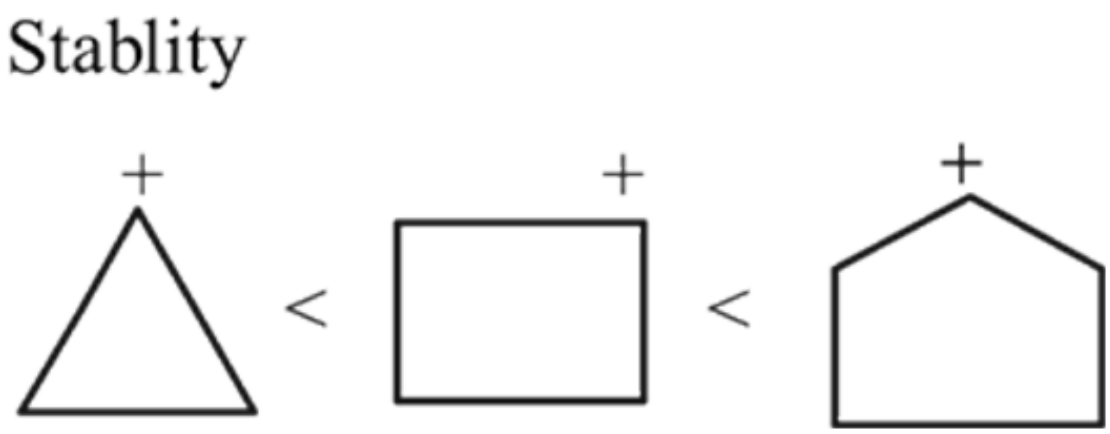
Solution:

4-nitrobenzyl chloride is likely to react by the S_N2 mechanism as the strongly electronwithdrawing nitro group would destabilize the carbocation intermediate of the S_N1 mechanism, a benzylic chloride that disfavours the S_N1 mechanism.

Q.20 Answer:

iii $>$ ii $>$ i

Solution:



@IITJEE_Advanced